

Computer Organization & Assembly Language (CSC-236)

Lecture 12

4

Bit Manipulations

4.2. SHIFTING AND ROTATIONS

- The set of shifting and rotation instructions is one of the most useful set in any processor's instruction set
- They simplify really complex tasks to a very neat and concise algorithm
- The shifting and rotation operations which are available in our processor are mentioned in the next slide

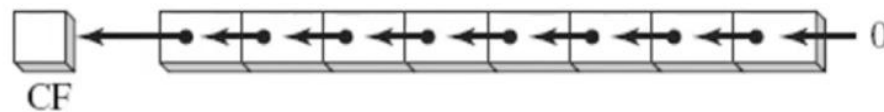
Shift and Rotate Instructions

SHL	Shift left
SHR	Shift right
SAL	Shift arithmetic left
SAR	Shift arithmetic right
ROL	Rotate left
ROR	Rotate right
RCL	Rotate carry left
RCR	Rotate carry right
SHLD	Double-precision shift left
SHRD	Double-precision shift right

3.) Shift Arithmetic Left (SAL)

SAL Instruction

- The SAL (shift arithmetic left) instruction works the same as the SHL instruction.
- For each shift count, SAL shifts each bit in the destination operand to the next highest bit position.
- The lowest bit is assigned 0. The highest bit is moved to the Carry flag, and the bit that was in the Carry flag is discarded.



4.) Shift Arithmetic Right (SAR)

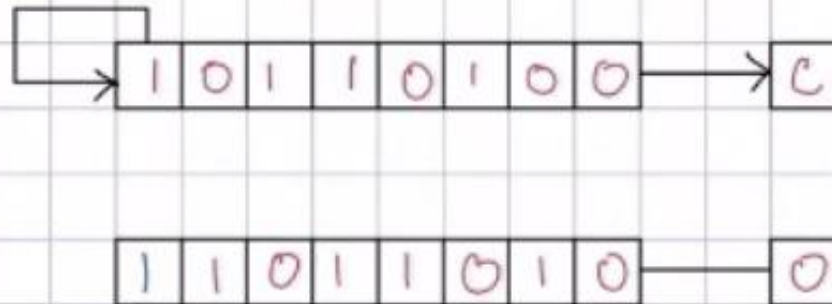
SAR Instruction

- The SAR (shift arithmetic right) instruction performs a right arithmetic shift on its destination operand.



4.) Shift Arithmetic Right (SAR)

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SAR Instruction: Example

- The following example shows how SAR duplicates the sign bit. AL is negative before and after it is shifted to the right.

```
mov al,0F0h           ; AL = 11110000b (-16)
sar al,1              ; AL = 11111000b (-8), CF = 0
```

4.) Shift Arithmetic Right (SAR)

SAR Instruction: Signed Division

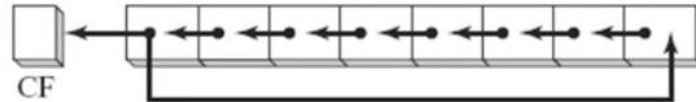
- You can divide a signed operand by a power of 2, using the SAR instruction.

```
mov dl,-128          ; DL = 10000000b
sar dl,3              ; DL = 11110000b
```


5.) Rotate Left (ROL)

ROL Instruction

- The ROL (rotate left) instruction shifts each bit to the left. The highest bit is copied into the Carry flag and the lowest bit position.
- The instruction format is the same as for SHL:



5.) Rotate Left (ROL)

Rotate Left (ROL)



5.) Rotate Left (ROL)

ROL Instruction: Example

- Bit rotation does not lose bits. A bit rotated off one end of a number appears again at the other end.

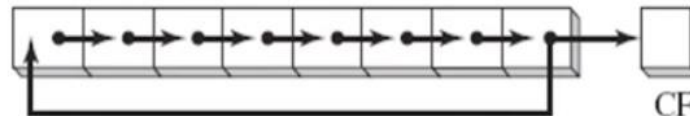
```
mov al,40h           ; AL = 01000000b
rol al,1             ; AL = 10000000b, CF = 0
rol al,1             ; AL = 00000001b, CF = 1
rol al,1             ; AL = 00000010b, CF = 0
```

```
mov al,00100000b
rol al,3             ; CF = 1, AL = 00000001b
```


6.) Rotate Right (ROR)

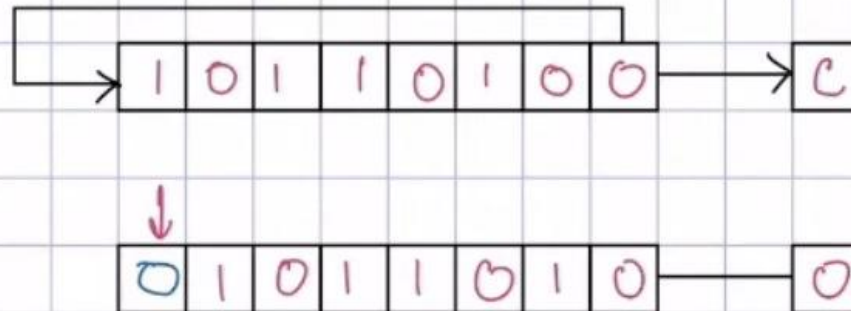
ROR Instruction

- The ROR (rotate right) instruction shifts each bit to the right and copies the lowest bit into the Carry flag and the highest bit position.
- The instruction format is the same as for SHL.



6.) Rotate Right (ROR)

Rotate Right (ROR)



6.) Rotate Right (ROR)

ROR Instruction: Example

```
mov  al,01h           ; AL = 00000001b
ror  al,1              ; AL = 10000000b, CF = 1
ror  al,1              ; AL = 01000000b, CF = 0

mov  al,00000100b
ror  al,3              ; AL = 10000000b, CF = 1
```


7.) Rotate Carry Left (RCL)

RCL Instruction

- The RCL (rotate carry left) instruction shifts each bit to the left, copies the Carry flag to the LSB, and copies the MSB into the Carry flag.
- If we imagine the Carry flag as an extra bit added to the high end of the operand, RCL looks like a rotate left operation.

<code>clc</code>	<code>; CF = 0</code>
<code>mov bl,88h</code>	<code>; CF,BL = 0 10001000b</code>
<code>rcl bl,1</code>	<code>; CF,BL = 1 00010000b</code>
<code>rcl bl,1</code>	<code>; CF,BL = 0 00100001b</code>

7.) Rotate Carry Left (RCL)

Rotate Through carry Left (RCL)



8.) Rotate Carry Right (RCR)

RCR Instruction

- The RCR (rotate carry right) instruction shifts each bit to the right, copies the Carry flag into the MSB, and copies the LSB into the Carry flag.



```
stc                ; CF = 1
mov ah,10h         ; AH, CF = 00010000 1
rcr ah,1           ; AH, CF = 10001000 0
```


8.) Rotate Carry Right (RCR)

Rotate Through carry Left (RCL)

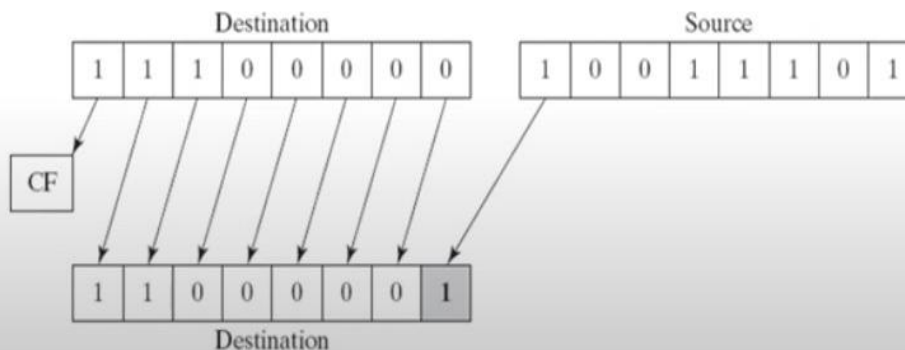


9.) Shift Left Double (SHLD)

SHLD Instruction

- The SHLD (shift left double) instruction shifts a destination operand a given number of bits to the left.
- The bit positions opened up by the shift are filled by the most significant bits of the source operand.
- The source operand is not affected, but the Sign, Zero, Auxiliary, Parity, and Carry flags are affected:

SHLD dest, source, count

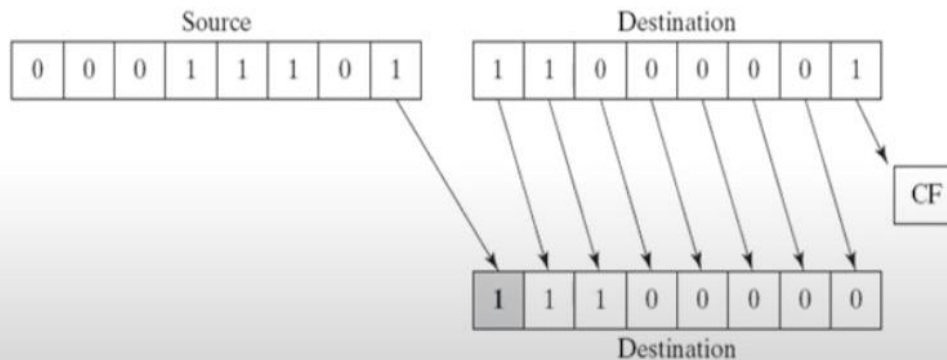


10.) Shift Right Double (SHRD)

SHRD Instruction

- The SHRD (shift right double) instruction shifts a destination operand a given number of bits to the right.
- The bit positions opened up by the shift are filled by the least significant bits of the source operand:

`SHRD dest, source, count`



Announcement :

“Quiz 3 in next class”

Date: (30-4-2025) / Wednesday

Syllabus : Chap 4 (Shifting and Rotatiion)

End of Lecture